

1. This is a quiz 4 question. Period is now 2π . Time in micro secs. See the Fourier series expansion in the quiz 4 solution.

$$V_g(t) = 25 + \frac{200}{\pi^2} \sin \omega t - \frac{200}{9\pi^2} \sin 3\omega t$$

$$N(s) \Rightarrow G(s) = \frac{1/LC}{s^2 + \frac{1}{RC}s + 1/LC}$$

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{8\pi \times 10^{-6}} = 10^6 \text{ rad/sec}$$

$$G(j\omega) \omega=0 = \frac{10^{12}}{(10^{12} - \omega^2) + j10^6 \omega}$$

DC $\omega=0$ $G(j0) = 1$

1st $\omega=10^6$ $G(j\omega) = -j1 \rightarrow 10^{12} - (10^6)^2 = 0$ $\frac{10^2}{j10^2} = -j1$

3rd $\omega=3 \times 10^6$ $G(j\omega) = \frac{1}{8+j3} = 0.117 \angle -159.44^\circ$

$$V_0 = 25 \times 1 + \frac{200}{\pi^2} \times 1 (\omega t - 90^\circ) - \frac{200}{9\pi^2} \times 0.117 \sin(3\omega t - 159.44^\circ)$$

$$= 25 + 20.26 \sin(\omega t - 90^\circ) - 0.2635 \sin(3\omega t - 159.44^\circ)$$

$$V_{rms} = \sqrt{25^2 + \left(\frac{20.26}{\sqrt{2}}\right)^2 + \left(\frac{0.2635}{\sqrt{2}}\right)^2}$$

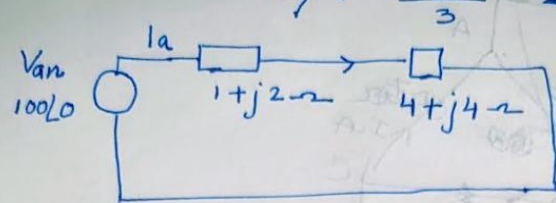
$$= 28.8144 \text{ V}$$

$$P_{max} = \frac{V_{rms}^2}{20 \times 10^3} = 41.52 \text{ mW}$$

- b) You can see $G(j\omega)$ is $-j1$ at the fundamental frequency ($\omega = 1 \text{ Mz}$)

Magnitude is 1 and phase is -90 degrees. Hence the gcf is 1 MHz and phase at this frequency is -90 degrees. You can also equate the magnitude to 1 and get the same answer.

2. 1- ϕ eqvt. circuit

$$Z_Y = \frac{12+j12}{3} = 4+j4$$


$$I_a = \frac{100}{1+j2+4+j4} = \frac{100}{5+j6} = 8.1967 - j9.8361 = 12.8 \angle -50^\circ$$

$$I_b = 12.8 \angle -170.19^\circ \text{ A} = -12.6118 - j2.1816$$

$$I_c = 12.8 \angle 69.81^\circ = 4.419 + j12.0172$$

$$W = V_{ac} \times I_a \cos \angle_{I_a}^{V_{ac}} \quad I_a = 12.8 \angle -50.19^\circ \text{ A}$$

$$V_{ac} = V_{an} - V_{cn} + I_c (1+j2)$$

$$= 100 \angle 0 - 100 \angle 120 + 12.8 \angle 69.81^\circ (1+j2)$$

$$= 146.077 \angle -26.75^\circ \quad [6]$$

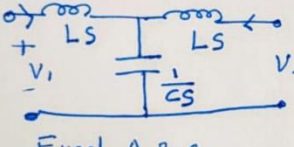
$$W = 146.077 \times 12.8 \cos(-23.43^\circ) = 1715.6 \text{ W}$$

you can also find out $V_{ac} = (1+j2) I_a + \frac{V_{ac}}{I_{ac} \cdot Z_L}$

(b) $V_{a2} = V_a - V_{a0} - V_{a1} = 1 \angle 60^\circ [1]$

$V_b = 3 \angle -120^\circ \quad V_c = 0 [1.5 + 1.5]$

3 (a) This is from the tutorial sheet.



$L = 47.75 \times 10^{-3} \text{ H} \quad S = j\omega \quad \omega = 2\pi f$

$f = 3000 \text{ Hz} \quad C = 0.2652 \times 10^{-6} \text{ F}$

Find A, B, C

Chara. imp. $Z_0 = \sqrt{\frac{B}{C}}$

Prop. constant $\gamma = \cosh^{-1}(A)$

$\cosh \gamma = A = 3.5$

$V_1 = A V_2 - B I_2$ Symmetric network

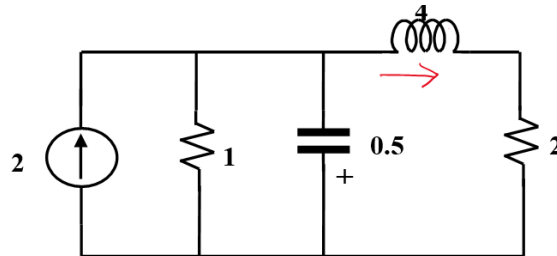
$I_1 = C V_2 - D I_2 \quad Z_{11} = Z_{22} = Z_0 \quad A = D$

$B = \frac{(1 - \omega^2 LC)^2 - 1}{j\omega C}$

$C = j\omega C \quad A = D = -3.5$

$Z_0 = j 670.83, \quad \gamma = 1.92 \text{ nepers}$

(b) The polarity of v_C and the direction of i_L are mentioned in the question. Not to change it.



Selecting capacitor voltage as x_1 (with polarity shown) and inductor current as x_2 (assuming the direction shown), KCL at A gives. Keep the identity of current source u

$$0.5\dot{x}_1 + x_1 + 2 - x_2 = 0 \quad \text{i.e., } \dot{x}_1 = -2x_1 + 2x_2 - 2 \quad (1)$$

$$\dot{x}_1 = -2x_1 + 2x_2 - 2u \quad (1)$$

$$x_1 + 4\dot{x}_2 + 2x_2 = 0 \quad \text{i.e., } \dot{x}_2 = -\frac{1}{4}x_1 - \frac{1}{2}x_2 \quad (2)$$

$$y = 2x_2 \quad (3)$$

$$A = \begin{bmatrix} -2 & 2 \\ -\frac{1}{4} & -\frac{1}{2} \end{bmatrix}, \quad B = \begin{bmatrix} -2 \\ 0 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & 2 \end{bmatrix}$$

eigenvalues are -1.5 and -1

eigen vector of $-1.5 \rightarrow \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ Eigenvector corresponds to $-1 \rightarrow \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

$P = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix} \quad A = P^{-1}AP = \begin{bmatrix} -1.5 & 0 \\ 0 & -1 \end{bmatrix}$

$\dot{z} = \bar{A}z + \bar{B}u$

$y = \bar{C}z$

$\bar{B} = P^{-1}B = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$

$\bar{C} = CP = \begin{bmatrix} 2 & 2 \end{bmatrix}$

4 (a) When a formal proof is asked where it mentions 'all the branch voltages' you have to prove

$V_b = Q^T V_t$ where V_b is the vector of all branch voltages. If you write, $V_c = -B^T V_t$, that is not correct. First part is a general proof. The second part is to demonstrate this proof for the graph given.)

4 (b) Eigenvalues 2 marks

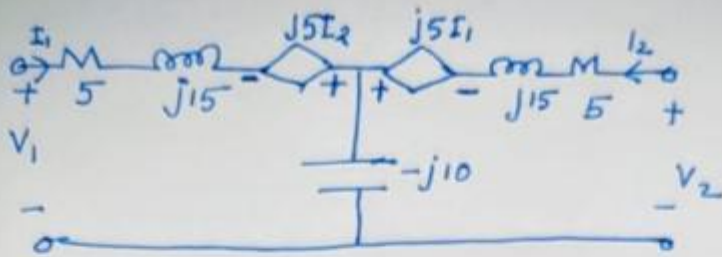
Response plot with justification. (Time constant, damping ratio, initial value, final value). 3 marks

The result of the code $x(m,:)$ 6 marks

5. There are two 2-port networks. Both are reciprocal and symmetrical. $A = D$ and $AD - BC = 1$

For the first network $A = 5$, $B = 24$; Hence $D = 5$; $AD - BC = 1$, hence $C = 1$

For the second network,



$$V_1 = AV_2 - BI_2 \quad \text{Find } A \text{ and } C \text{ with } I_2 = 0$$

$$I_1 = CV_2 - DI_2 \quad A = \frac{V_1}{V_2}, \quad C = \frac{I_1}{V_2}$$

From figure $V_1 = (5 + j15 - j10) I_1$ (because $I_2 = 0$) $V_1 = (5 + j5) I_1$
 $V_2 = -j5 I_1 - j10 I_1 = -j15 I_1$

$$A = \frac{V_1}{V_2} = \frac{5 + j5}{-j15} \quad C = \frac{I_1}{V_2} = \frac{1}{-j15}$$

$$A = D, \quad AD - BC = 1 \quad B = \frac{AD - 1}{C}$$

$$A = [0.333 + j0.333] \quad B = -3.33 + j15 \quad \frac{1}{3} \rightarrow \frac{1}{3}$$

$$C = j0.0667, \quad D = 0.333 + j0.333$$

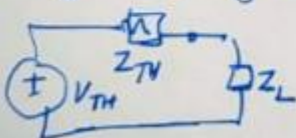
Cascaded network, $\begin{bmatrix} & \\ & \end{bmatrix} \times \begin{bmatrix} & \\ & \end{bmatrix}$

$$\begin{bmatrix} -1.6667 + j3.266 & -24.667 + j83 \\ -0.333 + j0.667 & -5 + j16.667 \end{bmatrix}$$

$$V_{TH} = \frac{V_g}{a_{11} + a_{21} Z_g}$$

$$Z_{TH} = \frac{a_{12} + a_{22} Z_g}{a_{11} + a_{21} Z_g}$$

$$V_g = 75, \quad Z_g = 1, \quad Z_L = 10$$



$$V_{TH} = -7.7037 - j15.1507$$

$$Z_{TH} = 23.1808 - j4.2445$$

$$I = \frac{V_{TH}}{Z_{TH} + Z_L}$$

$$V_2 = 5.081 \angle 70.34^\circ \text{ V}$$

$$V_2 = I_2 \times 10 = 1.7097 + j4.7848$$

$-100 + 4I + V_{TH} = 0 \quad (1)$
 $-100 + 4I + j10I + 5V_4 = 0 \quad (2)$
 $V_4 = 4I$
 $I = 5.3 \angle -148^\circ$
 Substituting in (2) $V_{TH} = 118.51 \angle 5.44^\circ V$
 Connect a Voltage source of 1V
 $I = I_1 + I_2$
 $I_1 = \frac{1}{4}$
 $I_2 = \frac{1}{8}$
 $V_4 = j10I_2 + 5V_4 \quad V_4 = 1$
 $Z_{TH} = \frac{1}{I_1 + I_2} = 1.124 - j.798$
 $R = |Z_{TH}| = 2.12 \Omega$
 $P_{max} = I^2 R = I \left(\frac{V}{Z_{TH} + R} \right)$
 $= 2164.10 W$

6a Maximum students attempted this problem. But in the last 2 steps made mistakes. R is the magnitude of Zth. In dc circuits, the voltage across R is Vth/2, but not in ac circuits.

Before, $t < 0$
 After, $t > 0$

$i_L(0) = \frac{12}{3} = 4 A$
 $V_C(0) = 8 V$

$-\frac{12}{5} + 1 \times I_1(s) + \frac{1}{s}(I_1(s) - I_2(s)) + \frac{8}{s} = 0$
 $(1 + \frac{1}{s}) I_1(s) + \frac{1}{s} I_2(s) = \frac{12}{s} - \frac{8}{s} = \frac{4}{s} \quad (1)$
 $I_2(s) + s I_2(s) - 4 - \frac{8}{s} + \frac{1}{s}(I_2(s) - I_1(s)) = 0$
 $-\frac{1}{s} I_1(s) + [1 + s + \frac{1}{s}] I_2(s) = 4 + \frac{8}{s} \quad (2)$

Solving (1) and (2) $I_2(s) = \frac{4(s^2 + 3s + 3)}{s(s^2 + 2s + 2)} = \frac{A}{s} + \frac{Bs + D}{s^2 + 2s + 2}$
 $A + B = 4$
 $2A + D = 12 \quad D = 0$
 $2A = 12 \quad A = 6$
 $B = -2$
 $D = 0$
 $I_2(s) = \frac{6}{s} - \frac{2(s+1)}{(s+1)^2 + 1}$
 $I_2(s) = \frac{6}{s} - \frac{2(s+1)}{(s+1)^2 + 1} + 2 \times \frac{1}{(s+1)^2 + 1}$
 $i_L(t) = 6 - 2e^{-t} \cos t + 2e^{-t} \sin t$
 $a \sin x + b \cos x = c \sin(x + \phi)$
 $c = \sqrt{a^2 + b^2} \quad \phi = \tan^{-1} b/a$
 $i_L(t) = 6 + 2\sqrt{2} e^{-t} \sin(t - 45^\circ)$

You can also solve the problem using state variable method. This problem was solved in the tutorial class, then some students commented that the inverse transformation is not possible. You can also do this by residue method.

Question 5

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a1=[5 24;1 5];
a11=(5+5i)/-15i;a21=1/-15i;
a22=a11;a12=((a11*a22)-1)/a21;
a2=[a11 a12;a21 a22];
a=a1*a2;% combined
am=abs(a);
ap=angle(a)*180/pi;
vg=75;zg=1;z1=10;
vth=vg/(a(1,1)+(a(2,1)*zg));
zth=(a(1,2)+(a(2,2)*zg))/(a(1,1)+(a(2,1)*zg));
i2=-vth/(zth+z1);
v2=i2*10;
i22 = -vg/((a(1,1)*z1)+a(1,2)+(a(2,1)*z1)+a(2,2));
% for checking
v22=i22*10;
% Some students converted ABCD parameters to Z
parameters
z11=a(1,1)/a(2,1);z12=1/a(2,1);z21=1/a(2,1);z22=a(2
,2)/a(2,1);
z=[z11 z12;z21 z22];
zthz=z22-((z12*z21)/(z11+zg));
vthz = (z21*vg)/(z11+zg);
i2z=-vthz/(zth+z1);
v2z=i2z*10;

```

ap =

3.6673	86.5878	117.0309	106.5514
0.7454	17.4005	116.5651	106.6992

Zm

4.9202	1.3416
1.3416	23.3452
Za 0.4658	-116.5651
-116.5651	-9.8658

th = 16.9968 -2.041